

Global fashion retail

How can sales data be used to understand demand drivers and support better stock planning decisions in a global fashion retail business

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1. Project overview
2. Dataset & methodology
3. EDA Global fashion retail business
4. Statistical validation
5. Forecast model & business recommendations
6. Tableau Dashboard
7. Conclusion



Project overview



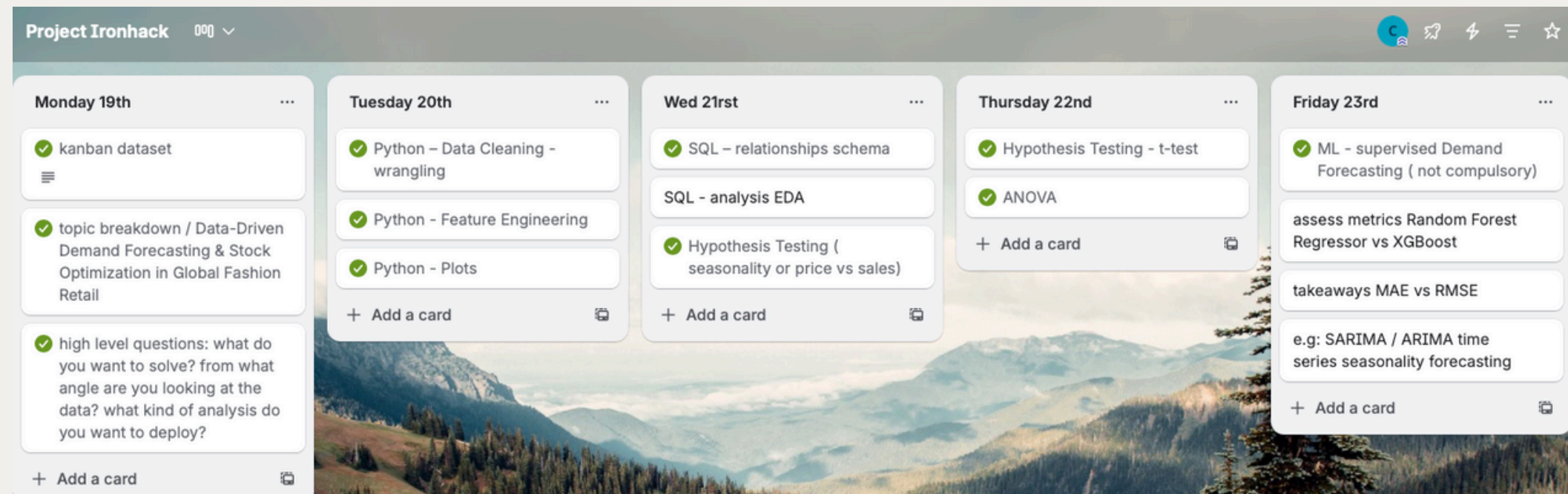
*This project analyzes sales data from **2023 to 2025** across Europe, China, and the United States to identify key demand drivers and support better stock planning decisions in a global fashion retail business.*

By applying exploratory data analysis, statistical testing, and machine learning techniques, the project examines how seasonality, product categories, pricing, and discounts influence demand patterns.

I chose this topic because I worked as a merchandise planner in the fashion industry dealing with challenges related to demand forecasting and demand drivers.

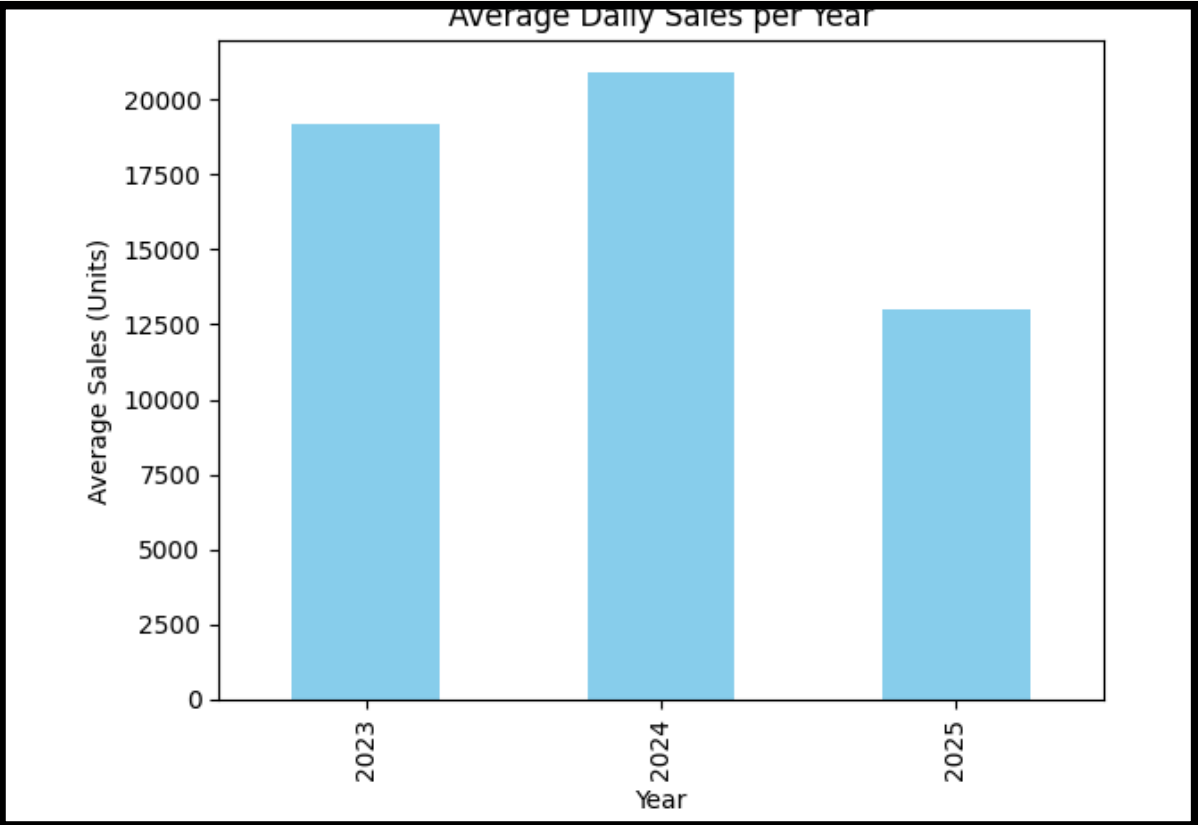
Methodology Framework

- **Project Management:**
Organized the workflow using a *Kanban* approach
- **Data Preparation**
Cleaned, merged, and processed six large datasets containing millions of records.
- **Data Wrangling:**
Standardized formats, created analytical features, and built aggregated datasets.
- **Exploratory & Statistical Analysis:**
Conducted EDA using Python (scatter plots, bar charts, time series) and performed hypothesis testing to answer key business questions.
- **Data Modeling (SQL):**
Designed relational schemas to support structured analysis.
- **Forecasting & Machine Learning:**
Applied *Prophet* for time series forecasting, including *train/test split*, *cross-validation*, and *performance evaluation*. Developed models at *product and category level* and defined improvement steps.
- **Data Visualization:**
Built *Tableau* dashboards to communicate key KPIs and insights to different stakeholders.

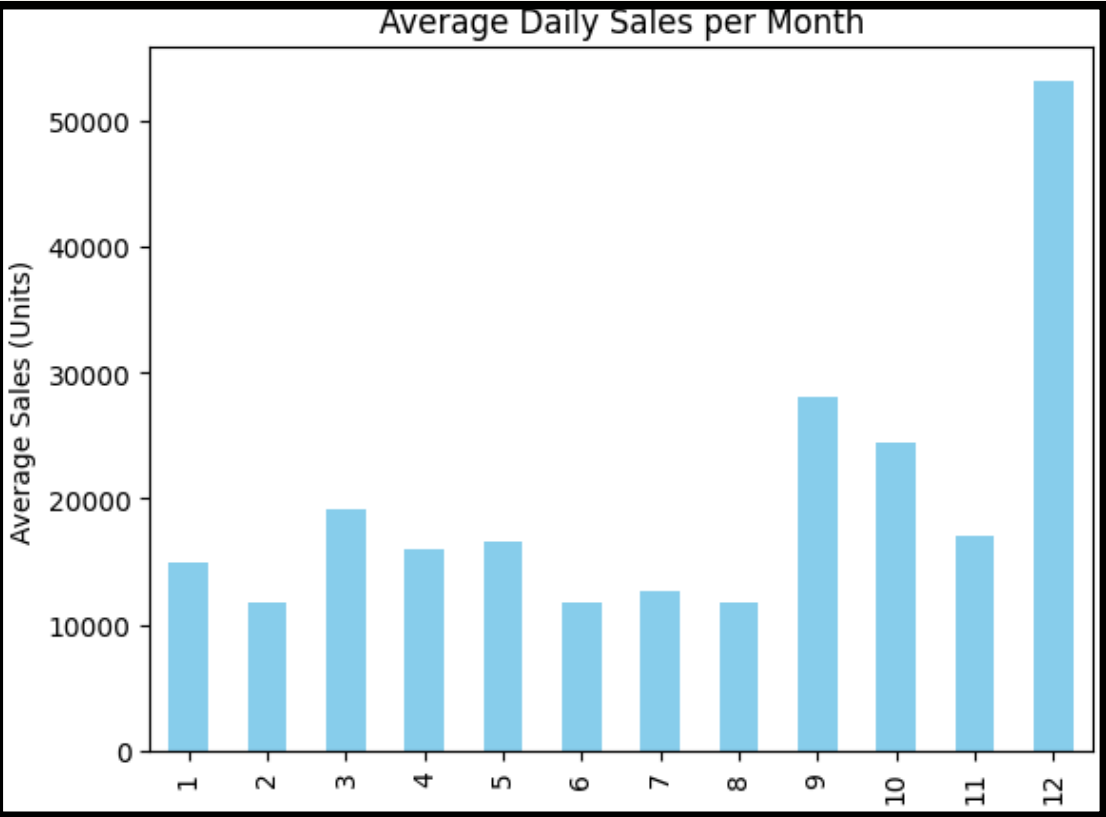


Q1.
How does
demand evolve
over time in
fashion retail?

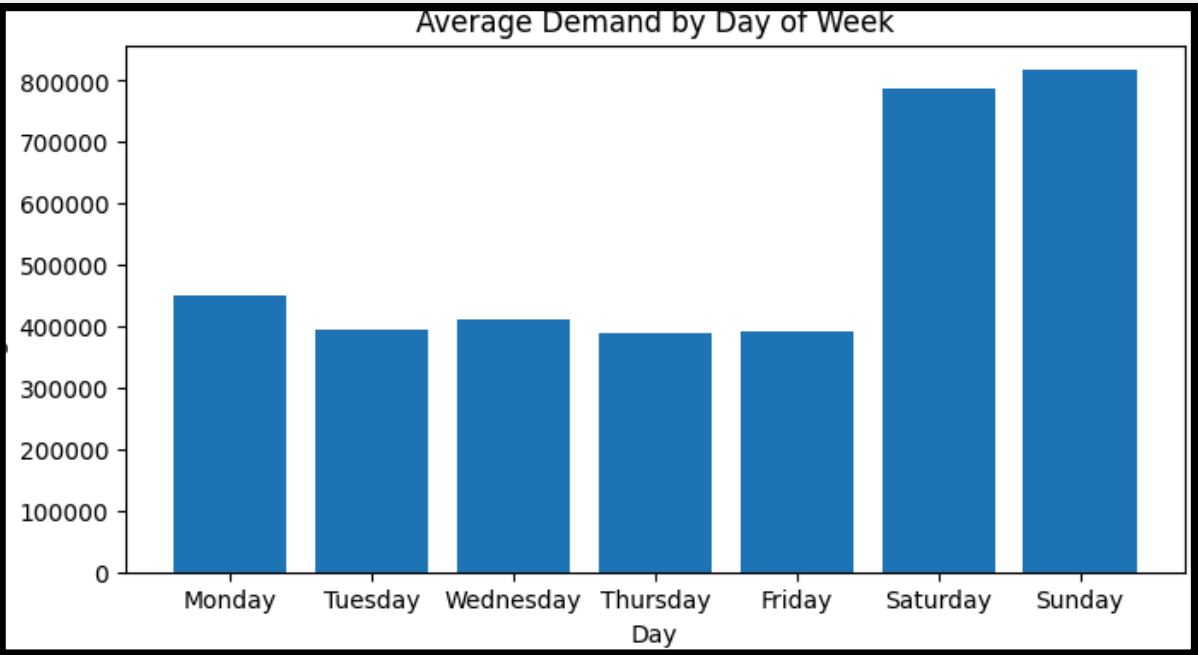
Yearly Trend



Monthly Trend



Weekly pattern



Stability/Volatility

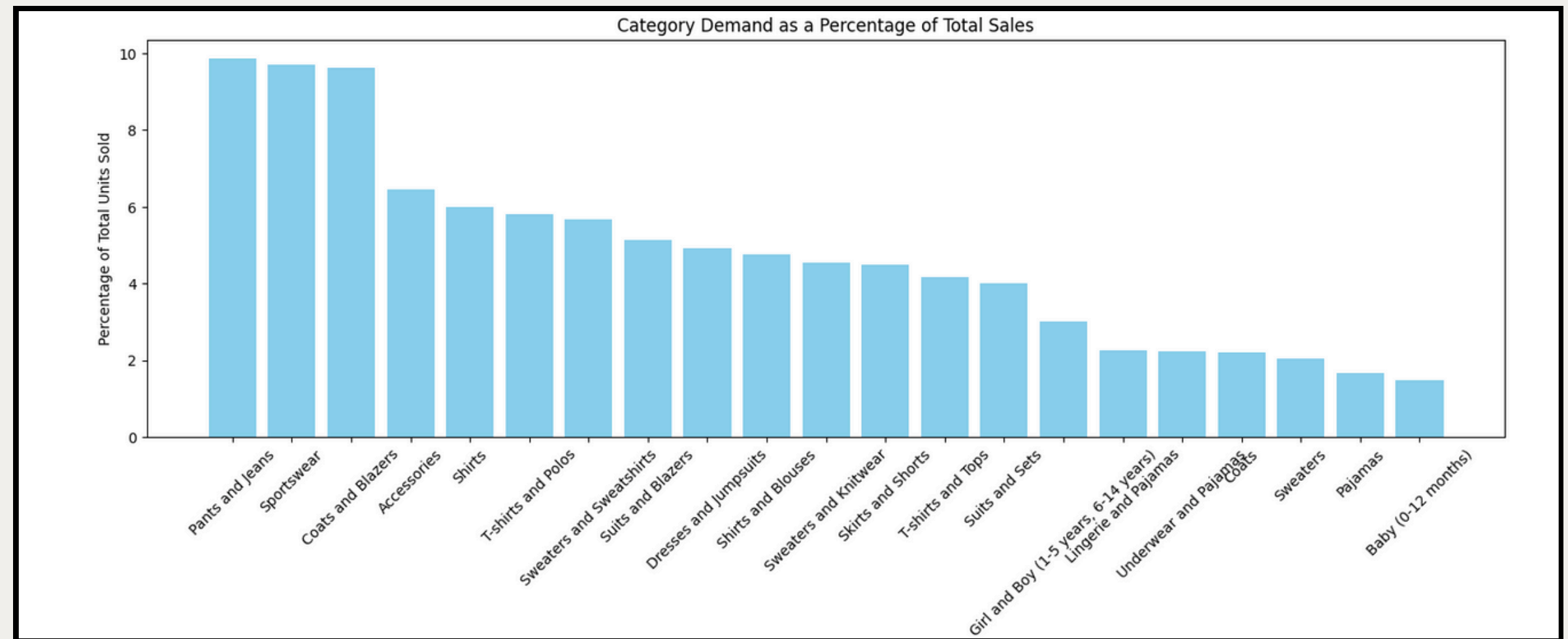
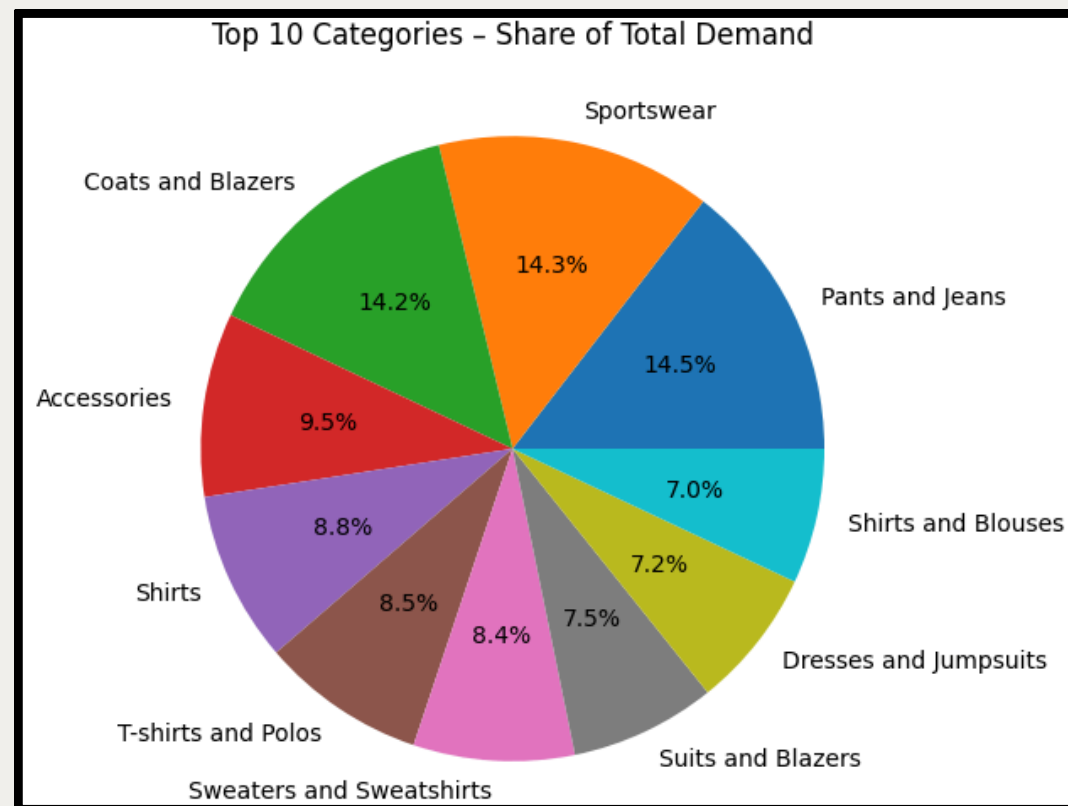
Mean = 520,290
Std = 194,008
CV = 0.37

Moderate volatility

Q2.

Where is the demand mostly concentrated ?

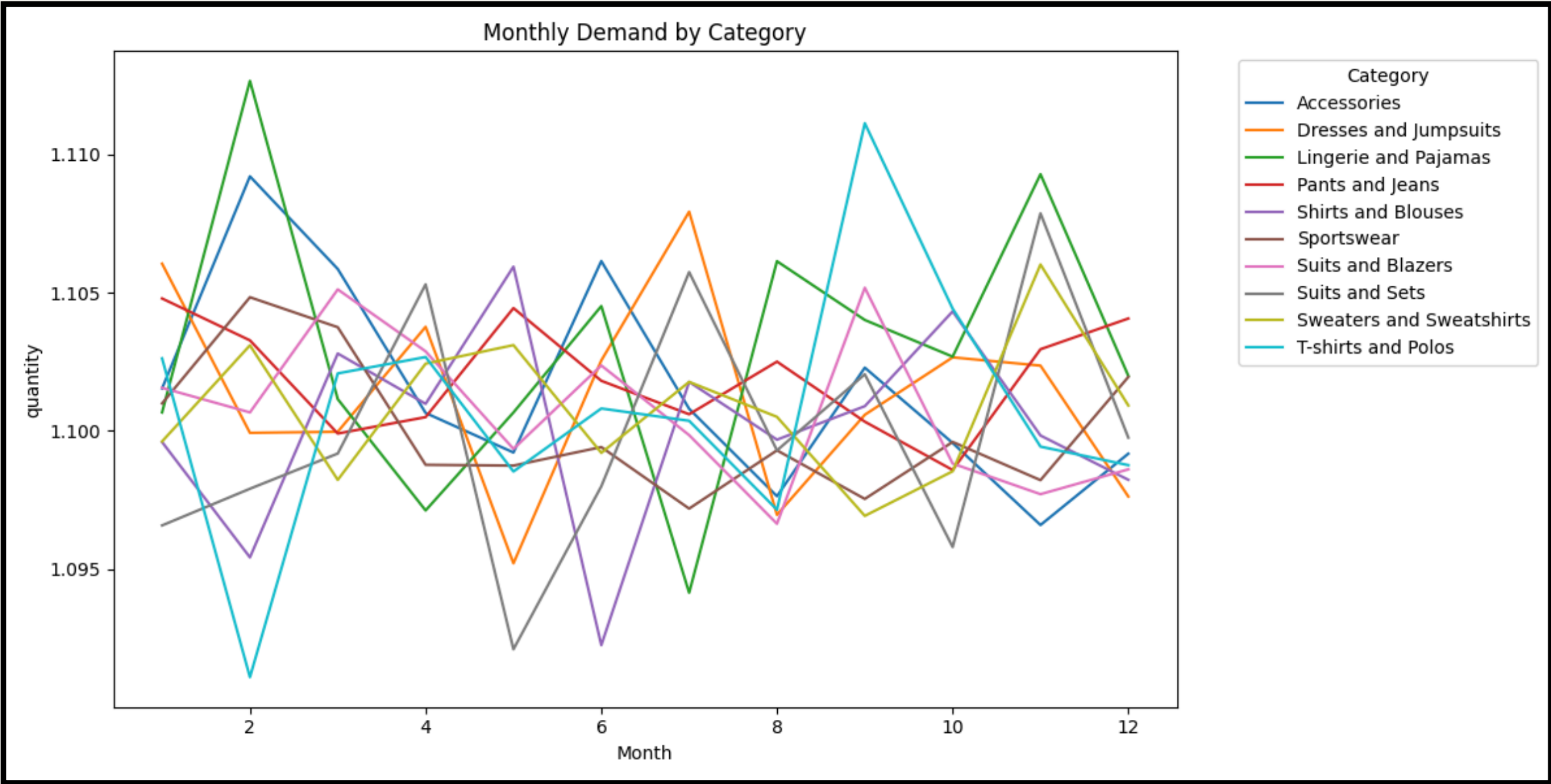
Category Drivers



Q2.

Where is the demand mostly concentrated?

CG contribution



Q3.

**Do Discounts
Really Increase
Demand?**

Full price share
unit sold

73%

Heavy markdowns
drive clearance, not
growth

Correlation -0.77
no linear relationship

**There is no
statistical evidence
that discounted
periods have higher
demand**

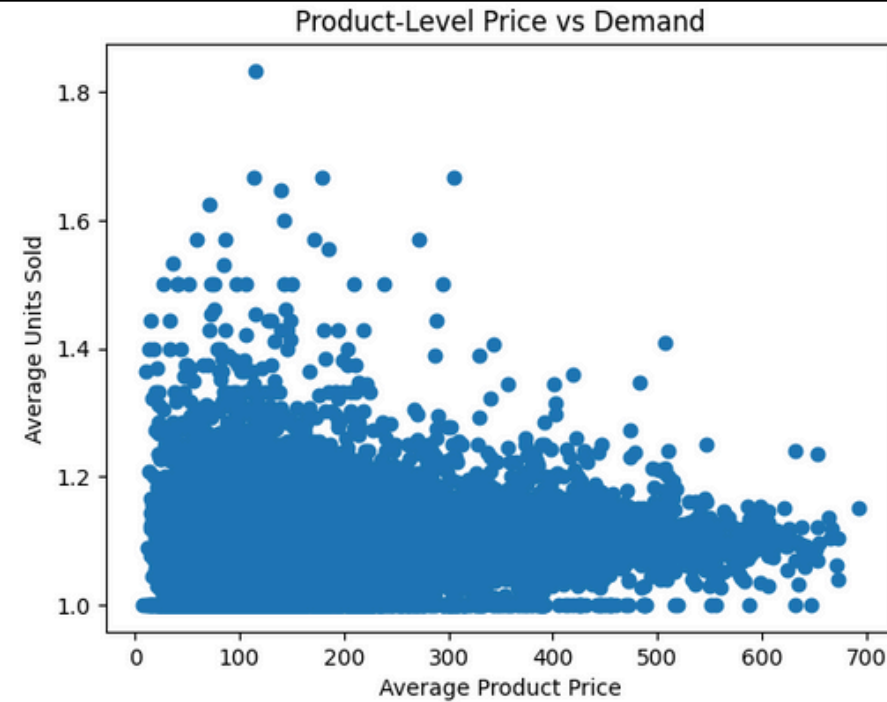
Test	P-value
t-test	0.99
Mann-Whitney	1

Discounts shift
demand between
products instead of
increasing total
volume

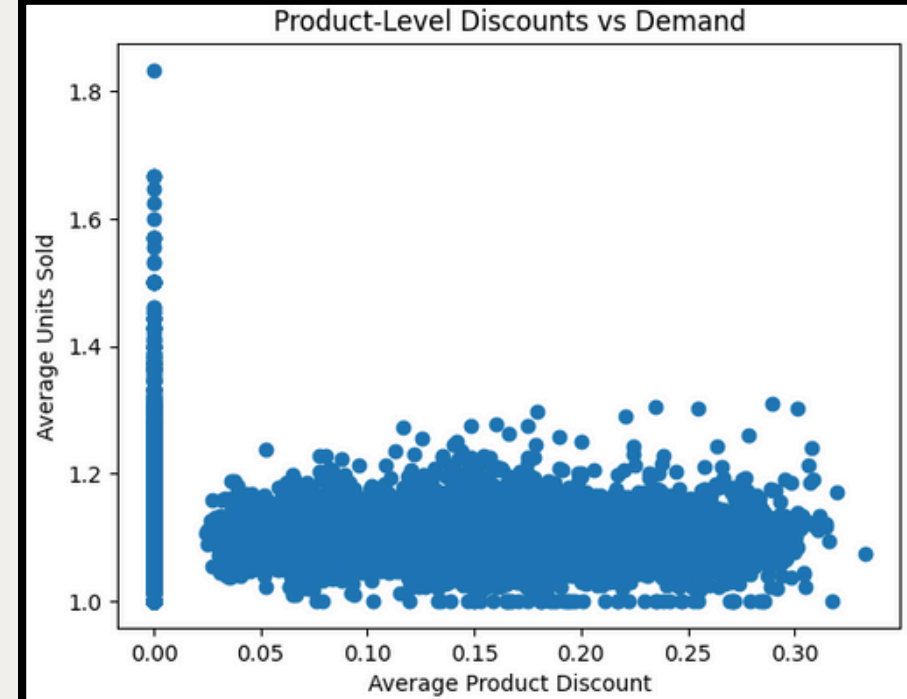
→
cannibalization risk

Q4. What drives demand variability in fashion retail?

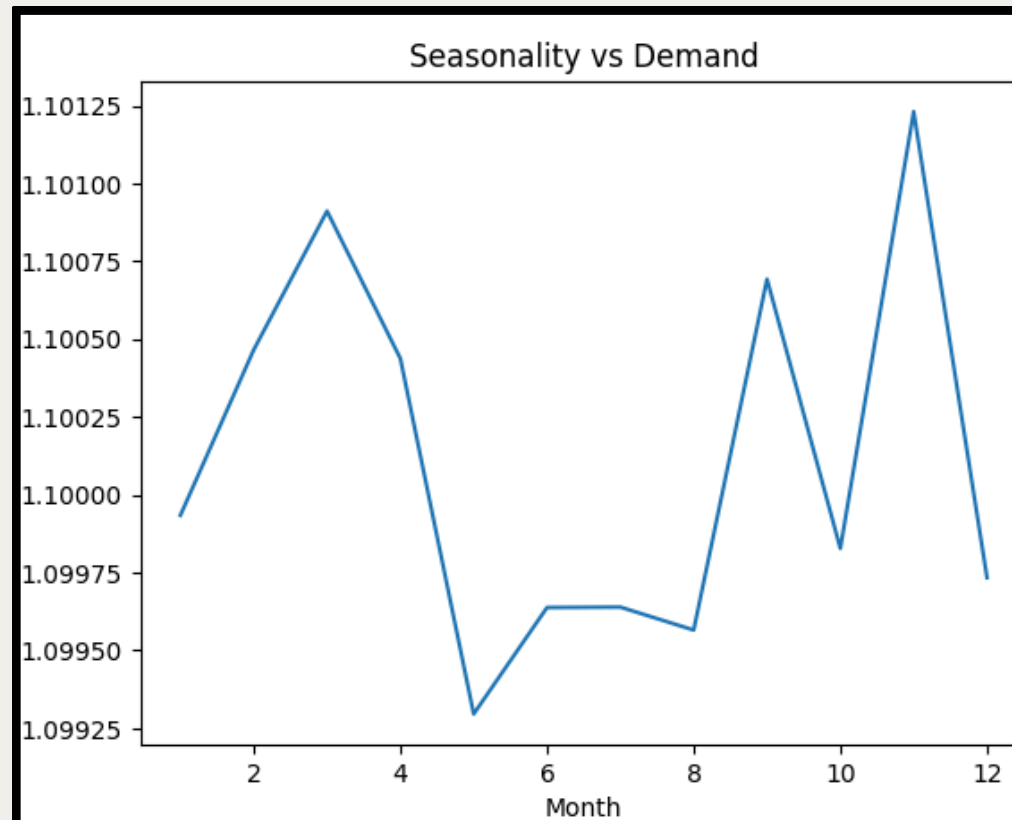
Price vs Demand



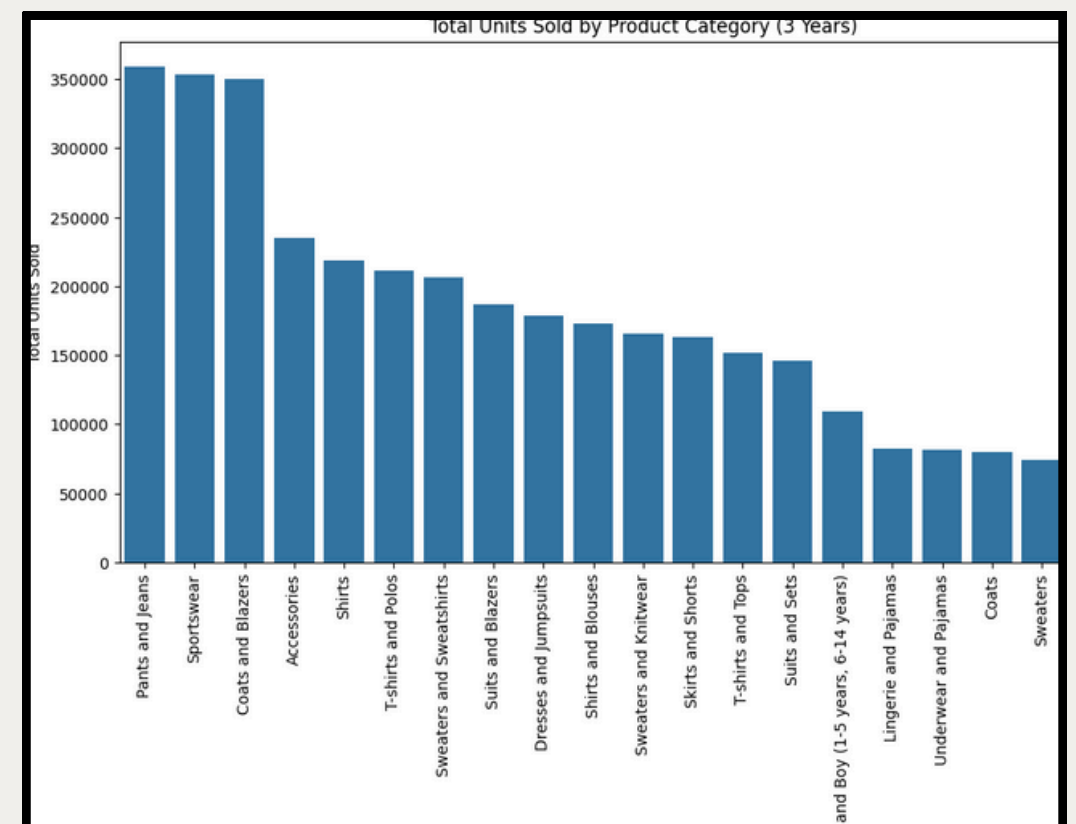
Discount vs Demand



Seasonality vs Demand



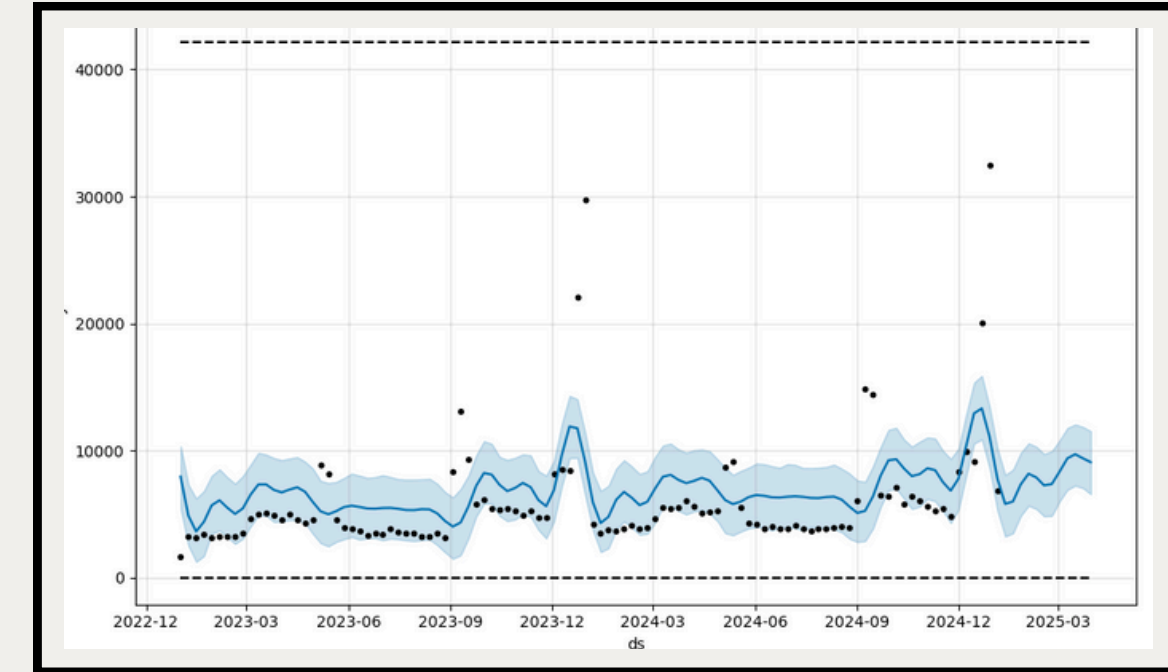
CG Demand



Demand Forecasting Methodology & Validation

Objective

Forecast *12-week* demand for Pants & Jeans category to support inventory planning.



Modeling Approach

Model: *Prophet*

Components:

Trend

Weekly & yearly seasonality

Discount as external regressor

Growth: Logistic

Baseline forecast
Unit_sold
103 K

SMAPE

Error range

23%

Bias

+1,500/week

Validation Strategy

Time-based Cross Validation

Rolling forecasts on Training sample:

Initial training: 12 months

Repeated every 3 months

Forecast horizon: 12 weeks

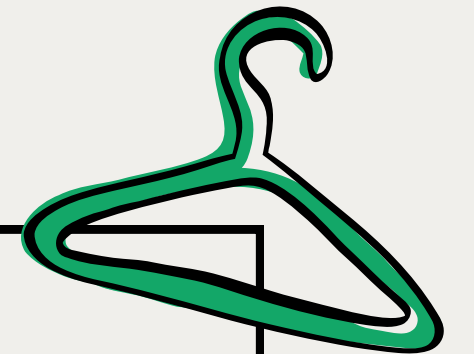
After bias correction, **expected demand** is around 85k units, with a range between 80k and 127k

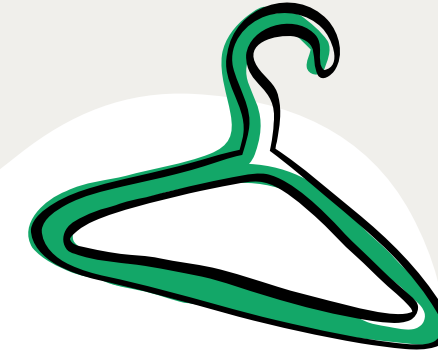
By progressively improving regressors, *tuning parameters*, and *validating performance*, the model can evolve into a robust decision tool for inventory management.

Tableau dashboard

Global fashion retail performance:

[https://public.tableau.com/app/profile/chiara.lanzi/viz/Global_Fashion_Retail_Performance/Globalsalesdemandoverview?
publish=yes](https://public.tableau.com/app/profile/chiara.lanzi/viz/Global_Fashion_Retail_Performance/Globalsalesdemandoverview?publish=yes)





The analysis revealed :

- clear seasonality effects and significant variations in demand between countries
- excessive discounting does not always lead to proportional sales growth.
- differentiated inventory strategies, with higher safety stock for fast-growing products and more conservative planning for minor categories.

This project showed:

- how historical sales data and business drivers can be used to build a reliable demand forecasting Prophet model for fashion retail.
- importance of model validation to offset overestimation bias



Thank you